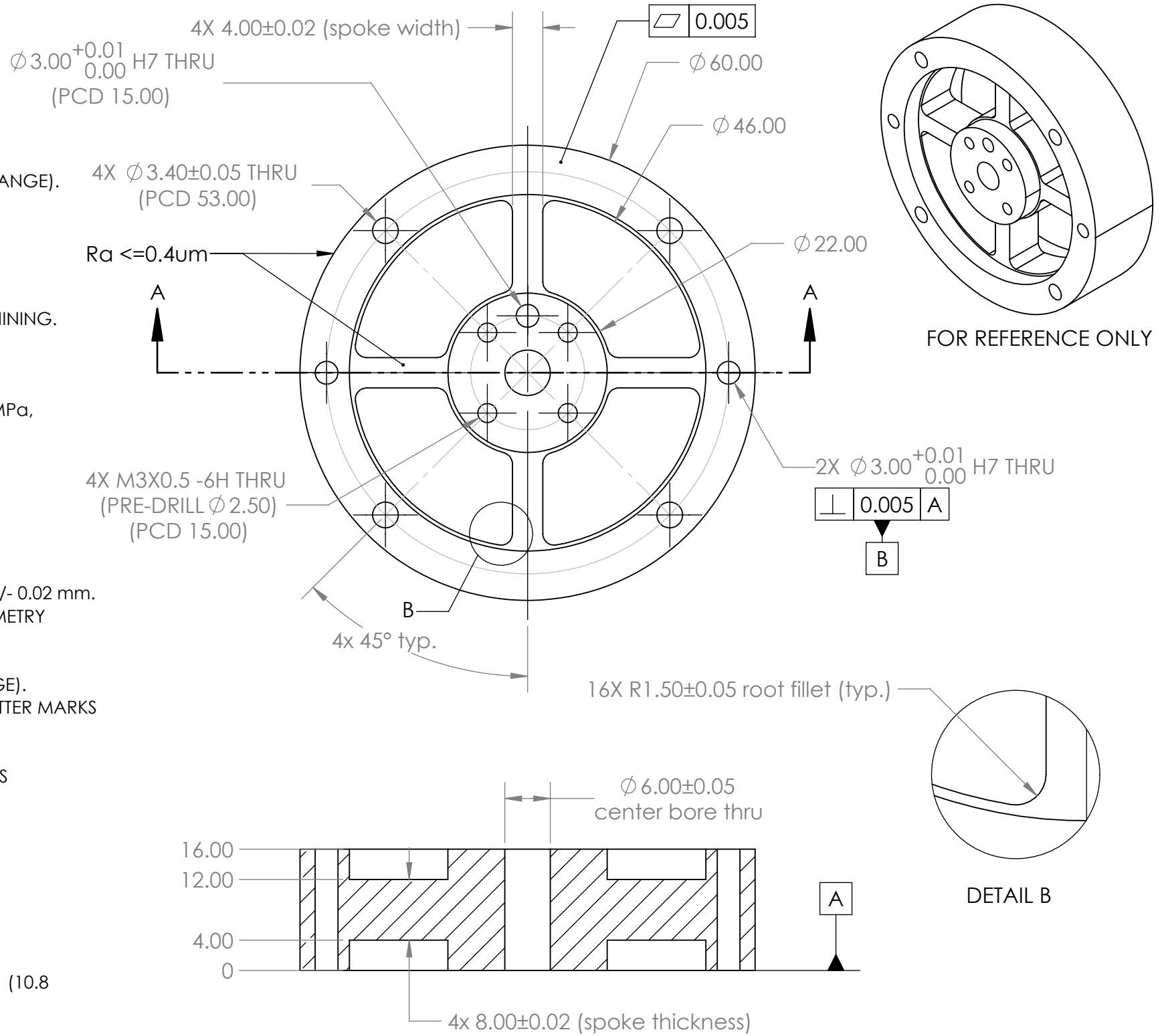


NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:
- 1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
 - 1.2 3D CAD DATASET (W3_ROBOT_6AXIS_RevA.STEP) GOVERNS NOMINAL GEOMETRY FOR ALL UNTOLERANCED SURFACES.
 - 1.3 GENERAL TOLERANCES: LINEAR +/- 0.1 mm, ANGULAR +/- 0.5 DEG (PER ISO 2768-m).
 - 1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS ARE IN MILLIMETERS.
 - 1.5 MOUNTING INTERFACE DESIGNED IN ACCORDANCE WITH ISO 9409-1 (MECHANICAL FLANGE).
2. MATERIAL & THERMAL HARDENING PROTOCOL:
- 2.1 PRIMARY MATERIAL: PRECIPITATION-HARDENED STAINLESS STEEL 17-4PH (UNS S17400 / 1.4542) PER ASTM A564 / AMS 5643.
 - CALCULATED FINISHED MASS: 185.1 g (VOLUME: 23,788 mm³).
 - 2.2 SUPPLY CONDITION: CONDITION A (SOLUTION ANNEALED) FOR ROUGH & FINISH MACHINING.
 - 2.3 PRECIPITATION HARDENING (CONDITION H900):
 - HEAT TREAT AFTER ROUGH MILLING: 480°C +/- 5°C (900°F) FOR 60 MINUTES, AIR COOL.
 - FINAL HARDNESS: HRC 40 TO 44.
 - MECHANICAL TARGETS: MIN YIELD STRENGTH Rp0.2 >= 1170 MPa, TENSILE Rm >= 1310 MPa, MODULUS OF ELASTICITY E = 196 GPa, POISSON'S RATIO = 0.272.
 - CONTRACTION ALLOWANCE: 0.05% TO 0.08% VOLUMETRIC CONTRACTION SHALL BE COMPENSATED PRIOR TO FINAL WIRE-EDM PASSES.
3. DfM SENSITIVE SPOKE & EDM FINISHING PROTOCOL:
- 3.1 CROSS-SPOKE GEOMETRIC SYMMETRY:
 - 4X ORTHOGONAL RADIAL SPOKES: WIDTH 4.00 +/- 0.02 mm, SENSITIVE THICKNESS 8.00 +/- 0.02 mm.
 - TOTAL AXIAL RECESS: 4.00 mm TOP AND 4.00 mm BOTTOM TO ENSURE COMPLETE SYMMETRY ABOUT Z=8.00 mm.
 - 3.2 STRESS CONCENTRATION ARRESTERS:
 - 16X R 1.50 +/- 0.05 mm INTERNAL SPOKE ROOT FILLETS (8X INNER HUB + 8X OUTER FLANGE).
 - TANGENTIAL BLEND MUST BE SMOOTH; NO UNDERCUTS, NOTCHES, OR CNC TOOL CHATTER MARKS PERMITTED.
 - 3.3 FINISH WIRE-EDM (WEDM) OPERATION:
 - 4X QUADRANT APERTURES AND SPOKE LATERAL SURFACES SHALL BE CUT VIA MULTI-PASS PRECISION WEDM AFTER H900 HEAT TREATMENT TO ELIMINATE RESIDUAL THERMAL DISTORTION.
4. STRAIN GAUGE INSTRUMENTATION & CALIBRATION THRESHOLDS:
- 4.1 STRAIN GAUGE BONDING AREAS:
 - 4X TOP AND 4X BOTTOM SURFACES OF SPOKES (Fz / Mx / My SENSITIVE).
 - 8X LATERAL FLANK SURFACES OF SPOKES (Fx / Fy / Mz SENSITIVE).
 - BONDING SURFACE ROUGHNESS MUST NOT EXCEED Ra 0.4 MICRONS.
 - 4.2 COMPATIBLE STRAIN GAUGES: TEMPERATURE SELF-COMPENSATING GAUGE TYPE STC-11 (10.8 ppm/°C).
 - 4.3 PERFORMANCE CRITERIA (NOMINAL RATED CAPACITY):
 - FORCE CHANNELS: Fxy = +/- 100 N, Fz = +/- 200 N.
 - TORQUE CHANNELS: Mxyz = +/- 10.0 N*m.
 - SENSITIVITY: 1000 +/- 200 MICRO-STRAIN AT FULL RATED LOAD.
 - CROSS-AXIS COUPLING ERROR (DECOUPLED): SHALL NOT EXCEED 0.5% FULL SCALE.
 - HYSTERESIS ERROR: <= 0.05% FULL SCALE.
5. SURFACE PASSIVATION, CLEANING & PACKAGING:
- 5.1 PASSIVATION: CHEMICAL CITRIC/NITRIC ACID PASSIVATION PER ASTM A967 (METHOD I OR II) TO RESTORE PASSIVE CHROMIUM OXIDE FILM POST-MACHINING.
 - 5.2 CLEANING: PRECISION ULTRASONIC DEGREASING IN ISOPROPYL ALCOHOL (IPA), DEIONIZED WATER RINSE, AND NITROGEN DRYING.
 - 5.3 PACKAGING: INDIVIDUALLY SEALED IN ANTI-STATIC, MOISTURE-BARRIER BAGS WITH DESICCANT.



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: ISO 2768-m (Medium LINEAR: ANGULAR:				FINISH: Ra <= 0.4 um (Spokes & Gauges), Ra <= 0.8 (Flange)		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION: A	
								ROBOT INTERFACE: ISO 9409-1 Flange Pattern Compatible			
								TITLE: ROBOTIC 6-AXIS F/T SENSOR ISOTROPIC ELASTIC BODY			
DRAWN								DWG NO.		A3	
CHK'D								W3_ROBOT_6AXIS_RevA			
APPV'D											
MFG											
Q.A						MATERIAL: 17-4PH (AISI 630) per ASTM A564 / AMS 5643 Age Hardened at 480°C ± 5°C (1 hr, Air Cool)					
						WEIGHT: 185.1 g (Net Monolithic Solid)		SCALE: NTS		SHEET 1 OF 1	